

# IT Report

## REPORT ETHICAL HACKING

METALBIN S.A. - METALCORP NETWORKS  
Date: 18/05/2026 | Andrew Alba | v1.0  
Type: Black Box | Printer exploitation and segmentation validation

### 1. DISCOVERED TOPOLOGY

Guest WiFi (192.168.10.100) → pfSense → LAN (hp JetDirect 9100 visible)  
DMZ (192.168.30.10) hidden (effective segmentation)

### 2. ASSET INVENTORY:

| ID  | Asset                                | IP             | Service        | Port      |
|-----|--------------------------------------|----------------|----------------|-----------|
| KL1 | Kali Linux (Guest)                   | 192.168.10.100 | -              | -         |
| PF1 | pfSense/Firewall                     | 192.168.10.1   | Firewall Web   | 80,443,53 |
| Z1  | Debian (Printer – hp JetDirect 9100) | 192.168.10.222 | CUPS/JetDirect | 631,9100  |

### 3. CRITICAL FIREWALL RULES ANALYSIS

- LAN Rules:
- Win7 OT, Windows 11, Servers isolated OK (rules #1-2)
  - Printer access (#5) ← Initial vector DMZ Rules:
  - Deny rest of DMZ → anywhere (effective)

## 4. DETAILED FINDINGS

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### **V-001 CRITICAL: Exposed pfSense Web Interface**

**IP: 192.168.10.1 | Services: 80/tcp nginx, 443/tcp nginx Vector: Direct access Guest WiFi (192.168.10.100)**

Evidence:

- HTTP/2 200 → pfSense Login Page confirmed.
- PHPSESSID cookies + CSRF tokens present.
- admin/pfsense → Blocked by CSRF (protection OK).
- The management interface is visible from Guest WiFi.

Severity: Critical (CVSS 9.1)

Impact: Total control of the infrastructure if valid credentials are obtained.

### **V-002 CRITICAL: CUPS Exploitation / Printer Access**

**IP: 192.168.10.222 | Service: CUPS 2.4 / cups-browsed**

**Initial access port: 631/tcp**

Status:

- Access to the printing service was confirmed.
- Post-exploitation was performed on the compromised host.
- Local system information was reviewed for access validation.

Impact:

- Compromise of the printing service.
- Possibility of obtaining sensitive local information.
- Foothold for persistence analysis and hardening.

### **V-003 CRITICAL: Privilege Escalation via User Artifacts**

**IP: 192.168.10.222 | File: ~/.bash\_history**

Detail:

- After initial access to the host, a review of the command history was performed.
- Useful artifacts for privilege escalation were identified.

Impact: • Elevated access to the system.

- Risk of exposure of credentials and sensitive files.

### **V-004 CRITICAL: Advanced Persistence (Systemd Backdoor)**

**Service: cups-maintenance.service**

## Status: Active / Persistent

### Detail:

- A persistent service was identified or implemented in /etc/systemd/system/.
- The service maintains continuous access to the compromised system.

### Impact:

- Persistence on the host.
- Risk of automatic reconnection after reboots.

## 5. RISK MATRIX

| Finding                                          | Severity | Likelihood | Impact | CVSS | Priority  |
|--------------------------------------------------|----------|------------|--------|------|-----------|
| V-001<br>pfSense(Router) Web                     | Medium   | Medium     | Medium | 7.5  | Medium    |
| V-002<br>Debian (Printer – hp<br>JetDirect 9100) | High     | High       | High   | 8.5  | High      |
| V-003 Escalation to<br>Root (Shadow file)        | CRITICAL | High       | High   | 9.8  | INMEDIATE |
| V-004 Persistence<br>(Backdoor)                  | CRITICAL | High       | High   | 9.0  | INMEDIATE |
|                                                  |          |            |        |      |           |

## 6. TECHNICAL RECOMMENDATIONS

- Limit printer access only to authorized hosts.
- Block or restrict the use of 631 and 9100 from Guest WiFi.
- Review and harden the CUPS service.
- Remove any unauthorized persistence in /etc/systemd/system/.
- Review ~/.bash\_history and credential handling policies.
- Maintain the hardening of the firewall/router.

## **7. EVIDENCE REFERENCE**

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- Day 1-2 Logs: full nmap / curl outputs.
- Evidence V-001: pfSense Login HTML + HTTP/2 Headers.
- Evidence V-002: access to CUPS service and post-exploitation.
- Evidence V-003/V-004: review of history, privileges, and persistence.

Status: Complete - Ready for IT implementation MetalBin S.A.